

# RAW SOCKETS

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## 1. CLDP Header Format [8 bytes header]

- Message Header: Stores 0x01, 0x02, 0x03 which tells which message type is sent. Size is 1 byte
- Payload Length: Store the message length sent in the data field. Size is 1 byte
- Transaction ID: A unique, random number. Size is 2 byte
- Reserved: Stores some extra information that user might want to send. I am using it to denote which Metadata client wants to ask. Size is 4 byte

## 2. Queries Supported

- Hostname (Reserved Field = 0)
- System Time (Reserved Field = 1)
- CPU Load (Reserved Field = 2)
- For all information at once, set the reserved field to any different number from above

## 3. Wireshark image + Terminal output

|             |                     |                 |      |                  |
|-------------|---------------------|-----------------|------|------------------|
| 1 0.000000  | 0.000000 127.0.0.1  | 255.255.255.255 | IPv4 | 57 Unknown (253) |
| 2 0.000165  | 0.000165 127.0.0.1  | 255.255.255.255 | IPv4 | 42 Unknown (254) |
| 3 10.001379 | 10.001214 127.0.0.1 | 255.255.255.255 | IPv4 | 57 Unknown (253) |
| 4 10.001541 | 0.000162 127.0.0.1  | 255.255.255.255 | IPv4 | 42 Unknown (254) |
| 5 12.050670 | 2.049129 127.0.0.1  | 255.255.255.255 | IPv4 | 57 Unknown (253) |
| 6 12.050990 | 0.000320 127.0.0.1  | 255.255.255.255 | IPv4 | 42 Unknown (254) |
| 7 20.002254 | 7.951264 127.0.0.1  | 255.255.255.255 | IPv4 | 57 Unknown (253) |
| 8 20.002734 | 0.000480 127.0.0.1  | 255.255.255.255 | IPv4 | 42 Unknown (254) |

```
Client: Waiting for HELLO...
Client: Received HELLO from server with IP 127.0.0.1
Client: Sending QUERY
Client: Waiting for RESPONSE...
Client: Received RESPONSE: Hostname: SUJAN-KGP, Time: 2025-03-31 15:09:15, CPU: 5.4% from IP 127.0.0.1
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Server: Sending HELLO
Server: Received packet (Type=0x02, TxnID=5678)
Server: Sending RESPONSE
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